H.W.

1. A study reports that the average systolic blood pressure of healthy adults is 120 mmHg with a standard deviation of 15 mmHg. A biologist measures the blood pressure of 40 students and finds a sample mean of 125 mmHg. At a 5% significance level, test whether the students have a significantly different average blood pressure from the general population.

2. A researcher wants to test if a new drug affects body temperature. The normal body temperature is 98.6°F. She records the temperatures of 10 patients after taking the drug and finds a sample mean of 99.1°F with a standard deviation of 0.5°F. Is there significant evidence at the 5% level that the drug changes body temperature?

3. A biologist is studying the presence of a genetic marker in frogs from two different ponds. The observed counts are:

	Marker Present	**Marker Absent**	**Total**
Pond A	25	15	40
Pond B	10	30	40

At a 5% significance level, test whether the distribution of the genetic marker is independent of pond location.

4. A biologist wants to compare the average heart rates of two species of frogs under the same laboratory conditions.

- For **Species A**, a sample of 60 frogs had an average heart rate of 75 beats per minute (bpm) with a known population standard deviation of 8 bpm.
- For **Species B**, a sample of 55 frogs had an average heart rate of 72 bpm with a known population standard deviation of 7 bpm.

At the 5% significance level, is there evidence of a significant difference in the average heart rates between the two species?

5. In a genetic study, the probability that a certain gene is present in an individual is 0.3. If 10 individuals are tested, what is the probability that exactly 4 individuals carry the gene?

6. A microbiologist counts the number of bacteria colonies forming on a petri dish. On average, 3 colonies form per dish. What is the probability that exactly 5 colonies will form on a randomly selected dish?

7. The body temperature of healthy rats is normally distributed with a mean of 37.5°C and a standard deviation of 0.5°C. What is the probability that a randomly selected rat has a body temperature between 37°C and 38°C?